

Database — Laboratory Work 1

ERD Diagram: International Airport Database

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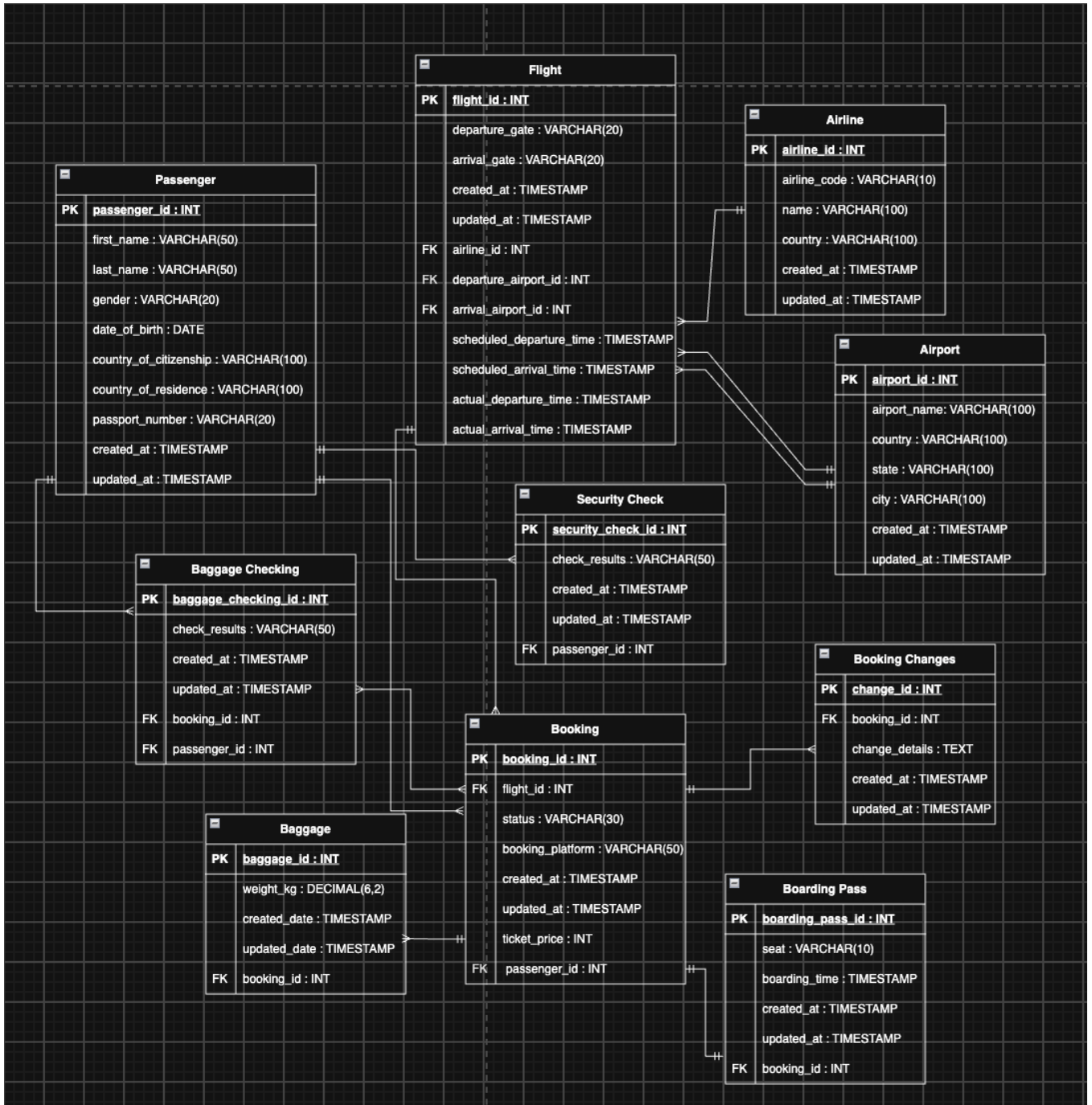


Figure 1. Entity-Relationship Diagram of the International Airport Database.

1. Legend

Symbol / notation	Meaning
PK	Primary Key — uniquely identifies a record in an entity.
FK	Foreign Key — references a primary key in another entity.
1:N	One-to-Many relationship.
1:1	One-to-One relationship.
	One side of a Crow's Foot relationship.
Crow's foot	Many side of a Crow's Foot relationship.

2. Entity Descriptions

Entity	Description
Airport	Stores information about airports: ID, name, country, state, city, creation date, and update date.
Flight	Stores scheduled and actual flight information, departure and arrival gates and times, airline ID, and departure and arrival airport IDs.
Airline	Stores airline information including ID, code, name, country, creation date, and update date.
Booking	Stores passenger flight bookings, including booking ID, flight ID, status, booking platform, timestamps, ticket price, and passenger ID.
Boarding Pass	Stores boarding pass information including ID, seat, boarding time, timestamps, and booking ID.
Baggage	Stores registered baggage information including ID, weight in kilograms, dates, and booking ID.
Baggage Checking	Stores baggage checking information including checking ID, results, timestamps, booking ID, and passenger ID.
Passenger	Stores passenger profile information including ID, name, gender, date of birth, citizenship, residence, passport number, and timestamps.
Security Check	Stores airport security check information including ID, results, timestamps, and passenger ID.
Booking Changes	Stores changes made to bookings separately, including change ID, booking ID, change details, and timestamps.

3. Relationship Descriptions

Relationship	Description
Airline — Flight	One-to-Many (1:N). A flight belongs to an airline through Flight.airline_id.
Airport — Flight (Departure)	One-to-Many (1:N). A flight references its departure airport through Flight.departure_airport_id.
Airport — Flight (Arrival)	One-to-Many (1:N). A flight references its arrival airport through Flight.arrival_airport_id.
Flight — Booking	One-to-Many (1:N). A booking references the flight through Booking.flight_id.
Passenger — Booking	One-to-Many (1:N). A booking references the passenger through Booking.passenger_id.
Booking — Boarding Pass	One-to-One (1:1) in this simplified model. The boarding pass references its booking through booking_id.
Booking — Baggage	One-to-Many (1:N). Registered baggage is associated with a booking through Baggage.booking_id.
Booking — Baggage Checking	One-to-Many (1:N). Baggage checking records are associated with a booking through Baggage_Checking.booking_id.
Passenger — Baggage Checking	One-to-Many (1:N). Baggage checking records reference the passenger through Baggage_Checking.passenger_id.
Passenger — Security Check	One-to-Many (1:N). Security check records reference the passenger through Security_Check.passenger_id.
Booking — Booking Changes	One-to-Many (1:N). A booking can have multiple recorded changes; each change references the booking through booking_id.

4. Design Notes

- The diagram uses primary keys (PK) and foreign keys (FK) to represent entity relationships and maintain referential integrity.
- Flight contains two separate references to Airport because an airport can be the departure airport for one flight and the arrival airport for another.
- The many-to-many relationship type is not explicitly shown in the provided schema; therefore, no direct M:N relationship is used in the diagram.
- Booking Changes is modeled as a separate entity because the assignment explicitly states that changes to a booking should be stored in a separate table.

5. Conclusion

In this laboratory work, an Entity-Relationship Diagram (ERD) for an International Airport Database was designed. The database structure includes the main entities required for managing airport operations, airlines, flights, passengers, bookings, boarding passes, baggage, baggage checking, security checks, and booking changes.

Primary keys were assigned to uniquely identify records in each entity, while foreign keys were used to establish relationships between related entities. The appropriate data types were selected for the attributes, and the relationships were represented using Crow's Foot notation with one-to-one and one-to-many cardinalities.

The designed ERD provides a clear and structured representation of how the main components of an international airport database are connected. It can serve as a foundation for implementing the database and maintaining consistent and organized information about flights, passengers, bookings, baggage, and airport operations.